PORTFOLIO

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General Problems in Mathematics

1. Problem in a Plane mirror Field-of-view solved by using complex numbers.

Consider a plane mirror of fixed length a. Suppose a point source O is placed in front of the mirror as shown in fig 1.1. It is also given that the ends of the mirror make angle α at O when subtended to meet. There is a straight screen which is placed to exactly fit the field of view of O w.r.t. the mirror. Furthermore, the ends of the screen when subtended to meet at the virtual image have equal lengths. Show that:

$$\text{Screen length} \propto \left|\sin\frac{\alpha}{2}\right|$$

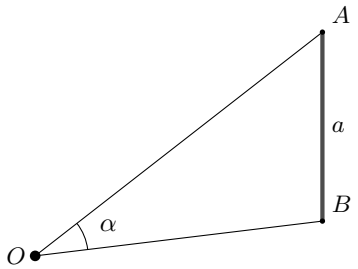


Figure 1.1: Geometric configuration of the mirror and the point source.

2.

We know that in the Binomial theorem for positive integral powers,

$$(1+x)^n = {}^nC_0 + {}^nC_1x + \cdots + {}^nC_nx^n, \qquad x \in \mathbb{R},\ n \in \mathbb{N}.$$

Similarly, let $W_n(x)$ be a function such that

$$W_n(x) = f^{(0)}(x) + {}^nC_1f^{(1)}(x) + {}^nC_2f^{(2)}(x) + \cdots + {}^nC_nf^{(n)}(x).$$

In a cleaner way,

$$\boxed{W_n(x) = {}^nC_0 f^{(0)}(x) + {}^nC_1 f^{(1)}(x) + {}^nC_2 f^{(2)}(x) + \cdots + {}^nC_n f^{(n)}(x)}$$

where $f(x)$ is an infinitely differentiable function over $x \in \mathbb{R}$ [i.e. differentiable of any order] and $f^{(r)}(x)$ denotes the rth derivative of $f(x)$; $f^{(0)}(x) = f(x)$.

For $W_n(x) = 0$, answer the following questions:

(i) Solve for $f(x)$.

(ii) What can be the degree of $f(x)$, if applicable?

(iii) How many roots can $f(x)$ have at most, and at least?

(iv) Suppose $S_i(x)$ is a function which is differentiable over $x \in \mathbb{R}$. Find the number of such $S_i(x)$ for which

$$\int_a^b f(x)\,\mathrm{d}x + \int_{f(a)}^{f(b)} S_i(x)\,\mathrm{d}x = b\,f(b) - a\,f(a); \qquad a, b \in \mathbb{R},\ a \neq b.$$

Assume that $f(x)$ is fixed w.r.t. $S_i(x)$.

3. Basic

Consider the following statements. For two differentiable functions $f, g : \mathbb{R} \to \mathbb{R}$,

S_1: There exists a c such that $c \in (a, b)$ and

$$\frac{g'(c)}{f'(c)} = \frac{g(b) - g(a)}{f(b) - f(a)}; \qquad f(b) \neq f(a).$$

S_2: There exists a $c \in (a, b)$ such that $g'(c) = \dfrac{g(a) - g(b)}{a - b}$.

S_3: There exists a $c \in (a, b)$ such that $f'(c) = \dfrac{f(a) - f(b)}{a - b}$.

S_4: If for $x \in [a, b]$, $f(a) = f(b)$ or $g(a) = g(b)$, then there exists a $c \in (a, b)$ such that $f'(c) = 0$ or $g'(c) = 0$ respectively.

[In all the above statements, $a, b, c \in \mathbb{R}$.]
Which of the following is/are correct?

(A) $S_1 \Rightarrow S_2$ or $S_1 \Rightarrow S_3$

(B) 'S_1 and S_2' $\Rightarrow S_4$

(C) S_2 and $S_3 \Rightarrow S_1$

(D) 'S_4' is derived from S_1, S_2, S_3 (any one of them).

4.

Let $I(x) = \int \frac{a(x)}{\big(b(x)\big)^2}\,\mathrm{d}x$, where $a(x)$ and $b(x)$ are real valued functions, $b(x) \neq 0$. Show that there exists a function $f(x)$, which is real valued, such that

$$\frac{f(x)}{I(x)} = b(x); \qquad I(x) \neq 0,$$

if not, then disprove it.

5. Consider a function $f(x) = {}^{n}C_{x}$ defined for $n \neq 0$, $0 \leq x \leq n$; $x, n \in \mathbb{Z} - \mathbb{Z}^{-}$ where $\boxed{{}^{n}C_{r} = \frac{n!}{(n-r)!r!}}$. In the XY-plane, if $f(x)$ is plotted on it then find the area under the path traced by connecting its points with a single straight lines (that is, connecting every two consecutive points with a single straight line).

6. Consider a function $f(x)$ such that

$$\sum_{n=1}^{k} f^{(4n)}(x) = 0,$$

where $f^{(r)}(x)$ denotes the r-th derivative of $f(x)$ and $k \in \mathbb{N}$. Then which of the following is/are true?

(A) All such $f(x)$ obeying the above equation are periodic.

(B) Infinitely many such $f(x)$ exist that satisfy the equation.

(C) The number of such $f(x)$ satisfying the equation is finite.

(D) There exist infinitely many $f(x)$ which are periodic and satisfy this equation.

7. Counting NENO combinations:

Let $f(x), g(x), h(x)$ be three real-valued functions, none of which is NENO (neither even nor odd) — that is, each of f, g, h is either even or odd. Using each of f, g, h at most once (no repetition), and combining them only with basic arithmetic operations and composition of functions, find the number of distinct functions $\alpha(x)$ that can be formed such that $\alpha(x)$ is NENO (neither even nor odd). Take f, g, h to be fixed with respect to $\alpha(x)$.

8. If $f : \mathbb{R} \to \mathbb{R}$ be an infinitely differentiable function such that

$$f(x) - 4f\left(\frac{x}{3}\right) + 2f\left(\frac{x}{a^2}\right) = 3x^2 \quad \forall x \in \mathbb{R},\ (a \in \mathbb{R},\ a \neq 0),$$

then:

(i) What is the nature of f (sinusoidal, exponential, polynomial, etc.)?

(ii) Prove the nature of f from the answer you got in (i).

(iii) If $f(\sqrt{8}) = 36$, then find all possible values of a^4.

(iv) Find all such possible $f(x)$.